

Where Workers Come From: An Occupational Flow Account for the United States, and the Instrument That Makes It Measurable

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Extended summary

An occupation's workforce is a stock fed by three inflows, people arriving from education, from other kinds of non-employment and from other occupations, and drained by two outflows, people leaving for another occupation and people leaving work. Workforce policy in the United States acts almost entirely on the first inflow, through the funding of training places, on the understanding that an occupation short of people is an occupation short of graduates. Whether that understanding is right depends on the relative size of the five flows, and this paper measures all of them, for every occupation the Current Population Survey can support, from a single survey in which each person is counted once on each side of the account.

Across the occupations covered, arrivals from another occupation run between four and twelve times arrivals from education, the lower figure across the twenty three major groups over the full period and the upper one at the detailed level in the most recent years. Arrivals from another occupation exceed all arrivals from outside employment in 170 of 172 occupations, the two exceptions both agricultural, and the ratio survives aggregation to the major groups, where the classification cannot confuse a nurse with a physical therapist. The arrivals are spread across the age distribution: for registered nurses the rate runs 3.1, 2.2, 3.4, 1.3 and 2.8 per hundred from age twenty five upwards, so they are not graduates recorded under the wrong heading, and lateral arrivals aged thirty five and over, taken on their own, outnumber every arrival from education by nearly seven to one. Nor are they people who earned a degree while working somewhere else: the share of lateral arrivals whose recorded education rose over the year is within two points of the share among people who stayed where they were.

Getting to that point required settling a measurement question first. Occupational mobility is usually measured by following a person across two interviews. The survey assigns occupation codes from written descriptions of what a person does, and two independent codings of a job the survey itself records as unchanged disagree in 48.1 per cent of cases at the detailed level and 31.8 per cent across the major groups, measured on 114,491 people who worked all fifty two

*Working paper. Every number, table and figure is produced by code held alongside the paper, which regenerates the whole of it from the public IPUMS extracts; the code is available on request. A companion paper takes the same account to the states.

weeks of the year. A linked measure of annual mobility returns 50.1 per cent, two points above that, so a panel applied to this survey recovers coding discordance with whatever real movement there is somewhere underneath. The March supplement instead asks in one interview what a person does now and what they did last year, both codes are assigned at one sitting from one description, and their errors cancel in the difference. That is why an instrument relying on recall outperforms a panel.

The account covers 230 occupations, annually from 2009 to 2025, by age band, with a standard error from the 160 replicate weights the Census Bureau publishes. Registered nurses lose 9.4 per hundred a year, a rate that among the large health and education occupations only physicians improve on, while nursing and home health aides lose 22.7, most of it to other occupations. Most movement between occupations also cancels: the change in occupation sizes that lateral movement produces is a tenth of the gross flow at the detailed level and less than that at minor and major group, so an occupation is churned by flows an order of magnitude larger than the change in its size.

1 Introduction

An occupation that cannot find enough people is usually treated as an occupation that has not trained enough of them, and states fund nursing school places, teacher preparation, apprenticeships and community college capacity on that understanding. The diagnosis has rarely been checked against a full account of where an occupation's people come from, because no such account exists at the level of detail policy operates on: employment statistics report stocks with precision and at high frequency, they say nothing about the flows that produced them, and an occupation stable in headcount is therefore recorded identically to one replacing a fifth of its people every year.

The account built here treats an occupation the way a demographer treats a population, as a stock observed at two dates with the arrivals and departures between them classified by where the person came from or went, and it is built from one survey in which each person is counted once on each side, so the identity closes without calibration to any external total.

Flows themselves have been studied for decades. Work on gross worker flows established that net employment change is the small residual of much larger movements between employment, unemployment and inactivity (Davis and Haltiwanger, 2014). Work on occupational mobility established that people change occupation often enough for occupation specific human capital to matter for wages (Kambourov and Manovskii, 2008), and that in the monthly Current Population Survey the measured rate depends heavily on whether the interviewer is allowed to carry last month's answer forward (Moscarini and Thomsson, 2007). Recent work treats transitions between occupations as a network with its own structure (Cheng and Park, 2020). None of it produces a table a workforce planner could open and read off, for a named occupation, how many people arrived from training, how many arrived from another occupation, how many left for another occupation and how many left work altogether.

Assembling one requires settling a measurement question the mobility literature has circled

for two decades. Kambourov and Manovskii (2008) devoted substantial effort to spurious transitions arising from independent coding, and Mellow and Sider (1983) showed much earlier that worker and employer reports of the same job frequently disagree. What has not been available is a direct measurement of the problem, on people whose occupation is known not to have changed. Section 2 provides one, Section 3 builds the account, Section 4 reports it, and Section 5 sets out the scope of the measures. The companion paper takes the same account to the geography and to the interstate movement of workers by occupation, and nothing here depends on it.

2 The instrument

2.1 Three measures of one transition

The Current Population Survey records occupation by asking what kind of work a person does, writing down the answer, and coding the description. Its March supplement asks that question twice, once about the present and once about the longest job held during the previous calendar year. The difference between the two codes is the annual transition used here.

The alternative is to follow a person across interviews, and the rotation design returns a household to the sample twelve months after its first interview, so the same person appears in March of one year and March of the next, with occupation collected at the time on both occasions and an eight month gap during which no answer is carried forward. Records are matched on the IPUMS person identifier and kept where sex agrees and age has advanced by no more than two years, the check IPUMS itself recommends. Linking the supplement to the March monthly file for the same month gives agreement on current occupation in 99.99 per cent of 504,531 people, so the two files share one coding of the present job. Table 1 sets the two measures of annual movement against a third: take last year’s occupation as recorded at the time, a year earlier, compare it with last year’s occupation as recalled in the supplement, and although both describe the same year they disagree 52.2 per cent of the time.

Table 1: Three measures of the same annual transition, per cent of linked people whose two codes differ, 2017 to 2025

	Detailed (474)	Minor group (97)	Major group (23)
Last year’s job, coded then and recalled now	52.2	44.0	33.5
A year apart, two independent codings	50.1	43.0	33.7
A year apart, both codes from one interview	10.3	6.7	3.3

The first two rows compare different things, March of one year against the longest job of that year in the first and March against March in the second, and neither is a measure of pure error, since real movement is present in both. What they share is two independent codings of one person’s work, and both sit near fifty per cent, so a linked measure of annual mobility in this survey lands at the level of the disagreement between two codings and cannot be separated from it. The third row is the supplement’s own measure on the same 146,558 people, and on the supplement as a whole it reads 12.2, 8.2 and 4.4 per cent across the three levels.

2.2 Which instrument is wrong

A disagreement establishes that one of two measurements is wrong without saying which, and two codes assigned at one sitting from one description could take the same value out of consistency alone.

Restrict attention to people whom the supplement records as having held the same occupation in both years and who worked all fifty two weeks. For these 114,491 people the independently coded occupation from a year earlier disagrees in 48.1 per cent of cases at the detailed level, 41.1 across the ninety seven minor groups and 31.8 across the twenty three major groups. The figure is an upper bound on discordance, since a person who changed occupation between March of the prior year and the longest job of that year sits in this sample with a real difference between the two codes, but the survey’s own annual mobility on the same linked people is 10.3 per cent at the detailed level and 3.3 at major group for a full year, and the window here is shorter than one. Real change of that order does not carry 31.8 per cent of people across a major group boundary. Figure 1 sets the three figures against the movement the supplement measures on the same people.

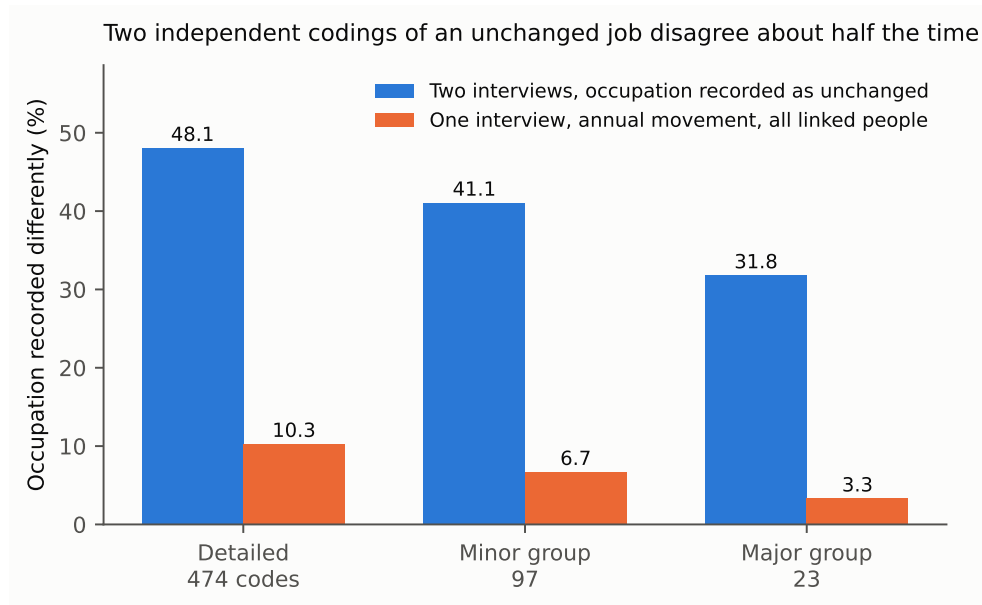


Figure 1: Two independent codings of an unchanged job disagree about half the time. The within interview measure records annual movement several times smaller at every level of aggregation.

Three things place the error in the coding. Sex and state of residence agree across the link by construction, since the sample is drawn by address and the match requires sex to agree, so the informative comparison is with the variables the link does not condition on: education, which people report afresh each time and which does change, differs in 12.6 per cent of cases, and occupation in 48.1, making occupation the least reliable variable in the survey by some distance. The disagreement sits between 47.4 and 49.6 per cent in all nine year pairs from 2017, including those either side of the survey’s 2020 change of occupation classification, which the harmonised codes carry across, and the pandemic years, when real movement did vary. The third concerns the shape of the disagreements, which would look like career moves if the supplement

were suppressing real ones, and do not. The largest single confusion is elementary and middle school teachers against secondary school teachers, at 0.96 per cent of all discordance, followed by first line supervisors against the staff they supervise in retail and in offices, by miscellaneous managers against specific managerial categories, by nursing aides against personal care aides, and by accountants against bookkeepers. Each is a boundary in the classification where a description could be coded either way.

The test bounds the disagreement between two independent codings of one job. It does not measure how far a single interview suppresses real change through the consistency of two codes assigned at one sitting, and the 12.2 per cent the supplement reports for the whole population should be read with that in mind.

Discordance concentrates within occupational families and along the supervisory boundary, and the pairs the test flags are the pairs the account should not report separately. Set a pair's discordance among unchanged jobs against the flow the supplement records between the two, both per hundred holders of either occupation. For registered nurses against physical therapists, a pair whose flow in the account looked implausible to me before this test, discordance runs 0.07 against a recorded flow of 0.89, so a second independent coding would add about a twelfth to the flow. For elementary against secondary teachers it runs 14.5 against 1.5, nearly ten times the flow, and for nursing aides against personal care aides 13.5 against 3.1. The teacher pair, and any pair like it, should be reported only at a level of aggregation containing both.

3 The flow account

3.1 Construction

The account uses the seventeen annual supplements from 2009 to 2025, as distributed by IPUMS (Flood et al., 2025), comprising 2,924,246 person records of all ages, of whom 2,231,260 are sixteen or over. Occupation is the OCC2010 variable IPUMS constructs, which carries the 2010 and 2018 Census classifications to one set of 474 codes, so the survey's change of classification in 2020 does not enter any comparison over time. A transition is the pair formed by a person's longest job of the previous calendar year and the job held in the survey week of March, so the interval it spans varies across people and a job held during the year that was neither the longest nor the current one is not seen. For each occupation and year the account records the number of people whose longest job of the previous year was in the occupation, the number arriving from education, from other non-employment and from another occupation, and the number departing to another occupation or out of employment. Arrivals from outside employment carry the previous year occupation code reserved for people who did not work at all, and the stated reason for not working separates education from caring, illness, retirement and inability to find work. An arrival from education is therefore a person who did no paid work in the previous year because they were studying; a student who held any job during the year is recorded under that job, and arrives in the account as a lateral mover or a stayer. Section 4.2 checks what that does to the figures.

The identity closes to zero people, since every person contributes once to each side from one source and no term is calibrated to an external total.

Two construction details matter for reading the figures. The 2014 supplement was fielded as a split sample, three eighths of respondents receiving redesigned income questions and five eighths the existing ones, and each half carries weights that sum to the full population, so analysing both halves together returns weighted totals twice the size of the population; filtering on the flag that marks the halves gives 144.8 and 145.5 million employed against 290.3 million unfiltered. Both halves sit on the trend between the 142.6 million of 2013 and the 147.8 million of 2015, and the account keeps the three eighths file, whose questions are the ones carried forward from 2015, with its own weights and replicates. Standard errors come from the 160 person replicate weights the Census Bureau constructs by successive difference replication (Fay and Train, 1995), with the multiplier of 4/160 the documentation prescribes (U.S. Census Bureau, 2024). Rates are recomputed inside each replicate, since numerator and denominator move together.

3.2 Coverage and precision

Precision degrades quickly as occupations get smaller. The median standard error on the rate of departure to another occupation is 1.3 percentage points where at least three hundred prior year holders are sampled, 2.8 points between one hundred and three hundred, 4.5 points between fifty and one hundred, and 7.1 points below fifty. A cell is reported only where at least one hundred prior year holders appear, a threshold set from that relationship before the estimates were inspected. That leaves 230 occupations reported in at least one year, 172 reported in at least one of the four years from 2022 to 2025, and 148 reported in all four, out of the 474 the survey distinguishes. The four year figures in Section 4 are the mean of the annual rates over the years in which an occupation clears the threshold, and their standard errors take the years as independent, which the overlap of rotation groups between adjacent years makes an approximation.

4 What the account shows

4.1 Arrivals

Table 2 gives the account for eight occupations, averaged over 2022 to 2025 and expressed per hundred people who held the occupation the previous year. Figure 2 shows arrivals and departures on a common scale.

Two features of the table cut against the way workforce shortage is discussed. Registered nurses lose 9.4 per hundred a year, and among the health and education occupations with a million or more holders only physicians, at 8.1, lose fewer; elementary teachers lose 10.3, postsecondary teachers 14.4, teacher assistants 16.8, and nursing and home health aides 22.7. Two thirds of the gap between nurses and aides is movement to other occupations, so what is discussed as a nursing shortage shows in the flows as turnover in the occupations next to

Table 2: Arrivals and departures per 100 prior-year holders, average of 2022 to 2025, standard errors in parentheses

Occupation	Arrivals from			Departures to	
	Education	Other non-work	Another occupation	Another occupation	Out of work
Registered nurses	0.7 (0.1)	0.7 (0.1)	3.4 (0.3)	4.4 (0.3)	4.9 (0.3)
Elementary and middle teachers	0.5 (0.1)	1.1 (0.1)	3.6 (0.3)	4.9 (0.3)	5.4 (0.3)
Secondary teachers	0.5 (0.2)	0.6 (0.2)	7.6 (0.8)	6.7 (0.7)	4.7 (0.6)
Software developers	0.4 (0.1)	0.5 (0.1)	4.1 (0.4)	6.7 (0.5)	5.5 (0.5)
Secretaries and administrative assistants	0.4 (0.1)	1.3 (0.2)	8.0 (0.6)	8.9 (0.5)	7.0 (0.5)
Truck drivers	0.4 (0.1)	2.1 (0.2)	7.7 (0.4)	9.3 (0.4)	9.6 (0.4)
Nursing and home health aides	0.9 (0.2)	1.8 (0.3)	13.1 (0.8)	13.2 (0.7)	9.5 (0.6)
Cashiers	4.8 (0.3)	3.0 (0.3)	8.7 (0.5)	12.3 (0.5)	20.3 (0.7)

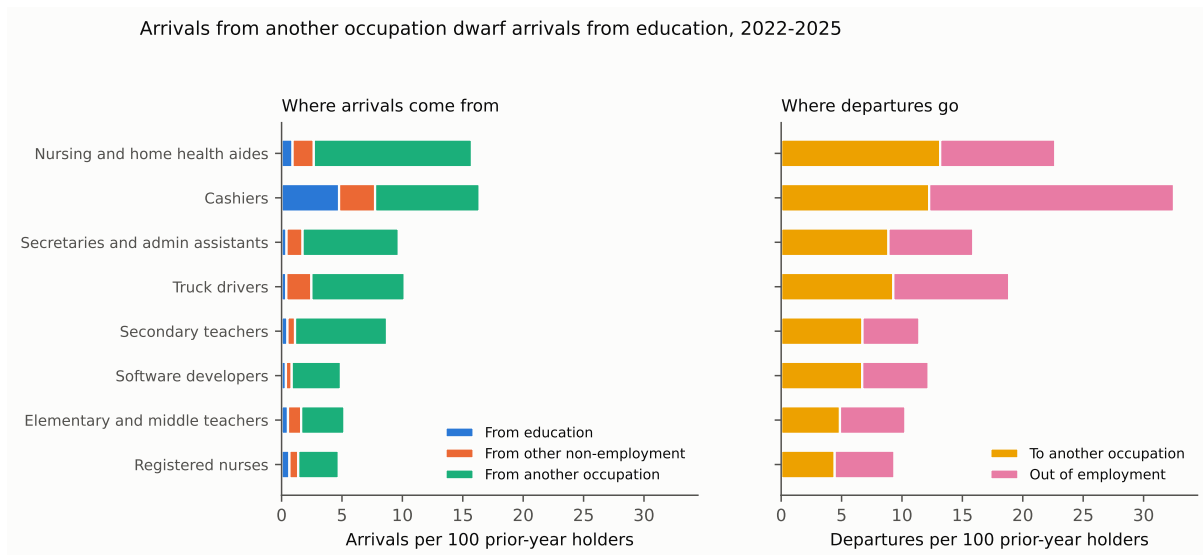


Figure 2: Arrivals from another occupation dominate arrivals from education, and the ranking by turnover runs against the way shortage is usually described.

nursing, and the aides take in almost as many people from other occupations as they lose to them. The second feature is the size of the lateral column. Education is the smallest of the three inflows in 83 per cent of the occupations reported, twenty three of them record no arrival from education at all across the four years, and the median occupation takes in nineteen people from another occupation for every one it takes from education, with the middle half of occupations lying between ten and thirty six.

The pattern extends across the whole account. Figure 3 plots every occupation reported, and 170 of the 172, or 99 per cent, sit above the line where arrivals from other occupations equal arrivals from outside employment; the two below it are farmers and ranchers, and miscellaneous agricultural workers, both of which draw on people who were out of work the previous year. Across the whole matrix, arrivals from another occupation run 4.1 times all arrivals from outside employment at the detailed level, 2.8 at minor group and 1.6 across the twenty three major groups, where the classification cannot confuse a nurse with a physical therapist and the surviving flow is movement between broad occupational families. Against arrivals from education alone the ratio is 12.1 at the detailed level and 4.7 at the coarsest, and over the full run from 2009 the corresponding figures are 10.6 and 4.0. The lowest the major group ratio falls in any single year is 3.1, in 2012, and the highest is 5.0, in 2024.

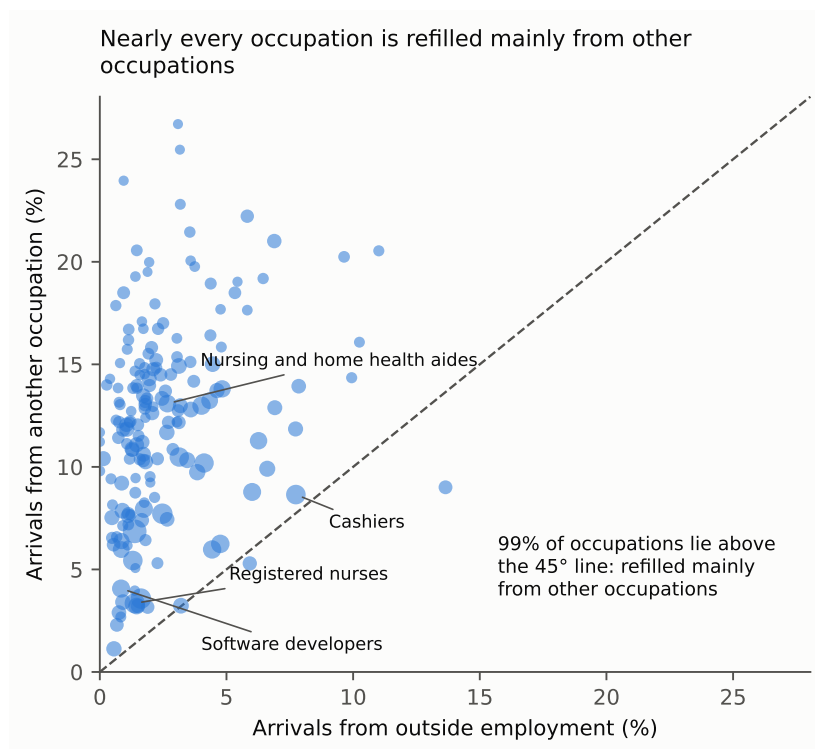


Figure 3: Arrivals from another occupation against arrivals from outside employment, one point per occupation, 2022 to 2025.

4.2 The age profile

Arrivals recorded as coming from another occupation might be graduates who held a job while studying, which would put the training pipeline back in the account under a different heading

and would concentrate such arrivals in the youngest bands. Figure 4 shows arrivals from another occupation by age for three occupations, and for registered nurses the rate runs 3.1, 2.2, 3.4, 1.3 and 2.8 per hundred across the bands from twenty five upwards. Between 42 and 51 per cent of lateral arrivals into the three occupations are under thirty five, against 35 per cent of the employed population, and across the whole matrix the figure is 43 per cent, where 92 per cent of arrivals from education are that young. The bound that matters is the one taken from the older bands alone: lateral arrivals aged thirty five and over, few of whom can be graduates under another heading, come to 6.8 times all arrivals from education at the detailed level, 4.4 at minor group and 2.2 at major group.

The other way training could sit in the lateral column is a person who earned a credential while working in a different occupation, the aide who becomes a nurse. The linked sample records education on both March visits, and the share of lateral arrivals whose recorded attainment rose over the year is 10.3 per cent against 8.5 among people whose occupation was unchanged, an excess of under two points, and 8.0 against 7.4 from age thirty five. The comparison registers a change in degree level only, so a certificate or a licence obtained while working does not appear in it, and attainment is recorded afresh at each visit, so the 8.5 among stayers is the noise in the recording; but a large flow of people arriving with a new degree would show as a large excess, and it does not.

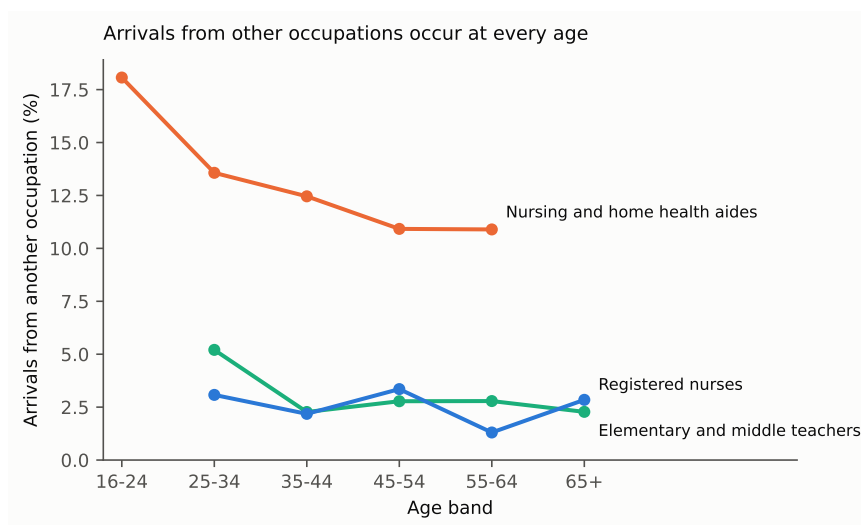


Figure 4: Arrivals from another occupation occur at every age.

4.3 Direction and churn

Gross movement between occupations is large and most of it cancels, and two measures say how much. The first sets, for each pair of occupations, the flow in one direction against the flow in the other and sums the absolute difference over pairs, the movement a pair does not return. The second takes each occupation's net lateral inflow, its arrivals from all other occupations less its departures to them, and sums the absolute values over occupations, halved so that each mover is counted once, the change in the size of occupations that lateral movement produces. The first bounds the second from above, since an imbalance on one pair can be undone around

a cycle of occupations. At the detailed level, gross lateral movement of 18.97 million a year over 2022 to 2025 contains 7.98 million of unmatched pairwise flow and 1.97 million of net change in occupation sizes, 42 and 10 per cent of the gross; at minor group the two shares are 26 and 7.5 per cent, and at major group 16 and 7.3. Over the full period from 2009 the occupation level share is 9.1, 7.8 and 7.7 per cent.

These are observed shares. Sampling error alone makes an absolute difference positive where the true flows balance, and coding discordance has no reason to be symmetric on every pair, since the two occupations in a pair differ in size and in how readily a description crosses the boundary between them, so neither figure is a floor on real reallocation. What the two say together is enough for the point: at every level of aggregation the change in occupation sizes produced by lateral movement is about a tenth of the movement itself, an occupation's stock is churned by flows an order of magnitude larger than the change in its size, and a policy raising arrivals works against departures of comparable magnitude.

5 Scope of the measures

Three features of the design bear on how the figures should be read. The prior occupation is the longest job of a calendar year and the current one is the job of a single week, so the count of people who held an occupation last year covers a twelve month window while the count of current workers is a moment, and the ratio between the two is not a growth rate; occupations also differ in seasonality and in how many jobs their people hold, and the longest job rule treats them differently in ways the account does not correct for. The survey looks back one year, so arrivals from outside employment include people returning to an occupation they had held before alongside those entering it for the first time. And a departure is the statement that a person whose longest job last year was in the occupation is not in it in the survey week, which a single transition cannot separate from a temporary absence.

6 Conclusion

Occupations in the United States are refilled predominantly by people arriving from other occupations, at four to twelve times the rate at which they are refilled from education. Arrivals from other occupations exceed arrivals from outside employment altogether in 99 per cent of occupations, the ratio survives aggregation to a level at which the classification cannot confuse adjacent professions and the removal of everyone under thirty five, and most of the movement cancels, leaving the change in occupation sizes at about a tenth of the gross flow.

Two independent codings of an unchanged job disagree in half of cases, so occupational mobility cannot be measured in this survey by following people between interviews, and the retrospective design, whose reliance on recall makes it look the weaker instrument, is the one that survives the comparison. I expected the opposite when I started, and the account would have been built on the linked panel had the test not been run first.

Training capacity, as far as this survey can see it, feeds the smallest of an occupation's

inflows. What the survey cannot see is a credential earned while working in another occupation, though the linked sample puts a low ceiling on how many lateral arrivals carry a new degree, and it is in the licensed occupations that such arrivals matter most. The largest inflow is governed by what determines who can move into an occupation from another one: credential requirements, licensure, and the transferability of skills between adjacent occupations. Which of those binds, and for which occupations, is the question the account was built to make askable, and the companion paper begins on it by asking where, within the country, the arrivals come from.

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